

Course Plan

Semester: 6

Year: 2025-26

Course Title: Generative AI	Course Code:25ECAC307
Total Contact hrs.: 70	Duration of ESA: 3 Hrs.
ISA Marks: 50	ISA Marks: 50
Lesson Plan Author: Dr. Uday Kulkarni	Date: 10-02-2026
Checked By: Dr. Meena S M	Date: 12-02-2026

Prerequisites: Basics of Python programming, Exploratory Data Analysis, Linear Algebra, Calculus, Statistical models and Machine learning concepts.

Course Overview:

This course introduces the fundamentals of Natural Language Processing (NLP), including applications, word embeddings, and parsing techniques such as dependency grammar and neural parsing. It covers sequence modeling approaches like Seq2Seq, attention mechanisms, autoencoders, and transformer architectures for machine translation and text generation. The course explores generative models including GANs, diffusion models, and their role in language and multimodal generation. It also provides an in-depth understanding of Large Language Models (LLMs) such as BERT and GPT, along with prompting and LoRA-based adaptation techniques. Students will learn multimodal AI concepts including vision-language models, CLIP, and text-to-image generation systems. Finally, the course introduces Retrieval-Augmented Generation (RAG), embeddings, vector databases, and context-aware AI system integration.

Course Outcomes (COs):

At the end of the course the student should be able to:

1. Apply NLP techniques such as word embeddings and dependency parsing for text understanding.
2. Implement Attention, and transformer models for machine translation and text generation.
3. Apply generative models such as GANs and diffusion models for language and image generation tasks.
4. Adapt and evaluate Large Language Models and multimodal models for NLP and vision-language applications.
5. Implement Retrieval-Augmented Generation (RAG) using embeddings, vector databases, and retrieval frameworks.
6. Develop a comprehensive project report and present research outcomes suitable for publication in a technical conference.

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Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)

Course Title: Generative AI	Semester: 6
Course Code: 25ECAC307	Year: 2025-26

Course Outcomes (COs) / Program Outcomes (POs)	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Apply NLP techniques such as word embeddings and dependency parsing for text understanding.	H	H										M	
2. Implement Seq2Seq architecture, Attention, and transformer models for machine translation and text generation.	M	M											H
3. Apply generative models such as GANs and diffusion models for language and image generation tasks.	H	M										M	H
4. Adapt and evaluate Large Language Models and multimodal models for NLP and vision-language applications.	M	M										H	H
5. Implement Retrieval-Augmented Generation (RAG) using embeddings, vector databases, and retrieval frameworks.	M	H										H	H
6. Develop a comprehensive project report and present research outcomes suitable for publication in a technical conference.		H						H	H	H			H

Degree of compliance **L**: Low **M**: Medium **H**: High

Competency addressed in the Course and corresponding Performance Indicators

Competency	Performance Indicators (PIs)	Course Specific performance indicators (CSPIs)
Competency 12.1: Demonstrate an ability to develop robust, scalable, and maintainable software systems using appropriate design principles.	PI 12.1.2: Construct a software system using a programming paradigm	12.1.2.1 Use programming languages like Python and frameworks such as TensorFlow, PyTorch, and NLTK for processing text data. 12.1.2.2 Apply python programming concepts in implementing, evaluating transformer networks and diffusion models for specific NLP tasks.
Competency 1.3: Demonstrate the competency in Engineering Fundamental	PI 1.3.2: Apply the principles of computer science including algorithms, data structures, and programming to develop a solution for complex engineering problems.	1.3.2.1 Apply appropriate data structures like graphs (for dependency parsing), trees (for syntactic parsing), and vectors for word embeddings. 1.3.2.2 Understand programming paradigms like object-oriented programming and functional programming helps optimize NLP algorithms. 1.3.2.3 Apply knowledge of machine learning fundamentals to understand and implement sequence-to-sequence (Seq2Seq) models and attention mechanisms, GAN, transformer-based models (like BERT, GPT) and language models.
Competency 2.1: Demonstrate an ability to identify and define complex engineering problems by understanding real-world contexts and constraints.	PI 2.1.1: Identify relevant problems with objectives by reviewing research literature.	CSPI 2.1.1.1: Identify Generative AI problems through research literature review. CSPI 2.1.1.2: Formulate problem objectives for NLP and multimodal AI applications based on literature.
Competency 2.2: Demonstrate an ability to apply principles of mathematics, natural sciences, and engineering to formulate and analyse complex problems, incorporating literature review to support sustainable development.	PI 2.2.2: Apply mathematical and scientific principles to formulate the engineering problem.	2.2.2.1 Use mathematical models such as vector space representation, cosine similarity, and matrix factorization in word embeddings. 2.2.2.2 Apply algorithmic knowledge like dynamic programming for dependency parsing while using Parsing techniques. 2.2.2.3 Exploring prompting strategies and fine-tuning techniques for specific NLP tasks. 2.2.2.4 Identifying limitations of existing LLMs and designing effective prompt engineering strategies.
	PI 2.2.3: Review and synthesize relevant research literature for problem formulation and solution approaches.	2.2.3.1 Analysis of LSTMs, GRUs, GAN, transformer and language models. 2.2.3.2 Evaluate pros and cons of using LSTMs, GRUs, GAN and Transformers for translation.
Competency 13.1 : Develop AI-based models to pre-process, analyse, and extract insights from structured and unstructured data.	PI 13.1.1: Build AI/ML models for data engineering tasks.	13.1.1.1 Implement mathematical models for autoencoders and decoders in NLP tasks such as text compression and summarization. 13.1.1.2 Designing attention-based models to enhance machine translation accuracy. 13.1.1.3 Apply machine learning principles to Implement various mathematical models like DC

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		GAN, Style GAN and Cycle GAN for image generation. 13.1.1.4 Applying probabilistic models to diffusion-based text generation. 13.1.1.5 Building and fine-tuning pre-trained LLM models for NLP tasks.
Competency 5.2: Demonstrate an ability to select and apply domain-specific tools, modelling techniques, and resources.	PI 5.2.2: Apply domain-specific tools and utilize computing platforms, libraries, and resource to solve engineering problems.	5.2.2.1 Implementing using LSTMs, GRUs, GAN and Transformers using frameworks like TensorFlow, PyTorch, and Transformers. 5.2.2.2 Running experiments on cloud-based AI platforms such as Google Colab. 5.2.2.3 Fine-tuning Gen AI models for domain-specific applications like medical text generation. 5.2.2.4 Adapting Gen AI models for controlled text and imagen generation. 5.2.2.5 Using OpenAI API, , and Google Vertex AI for LLM for text and image-based applications.
Competency 9.2: Demonstrate an ability to communicate effectively to various engineering community and society.	PI 9.2.1: Communicate effectively across cultural, linguistic, and disciplinary boundaries within teams and the broader society.	9.2.1.1 Students form diverse project teams to analyze real-world NLP applications such as sentiment analysis, chatbot design, and translation systems. 9.2.1.2 Teams collaboratively explore use cases of BERT and GPT and other models in finance, healthcare, law and education domains. 9.2.1.3 Team members contribute based on individual strengths—model design, dataset preparation, coding, testing, and documentation.
Competency 8.1: Demonstrate an ability to contribute effectively as a team member and lead diverse/multidisciplinary teams.	PI 8.1.2: Participate in team discussions, contribute ideas, provide feedback, and critique peer inputs.	8.1.2.1 Teams jointly implement Gen AI models and integrate modules through workflows. 8.1.2.2 Collaborative research presentations on Gen AI models and their applications. 8.1.2.3 Each member is responsible for presenting a specific part of the model or experiment during team evaluations.
Competency 9.1: Demonstrate the ability to comprehend, write, and present technical information for diverse audiences.	PI 9.1.2: Organize and produce well-structured content in a document or presentation with a clear and logical flow.	9.1.2.1 Students use tools like draw.io, or overleaf to create well-labelled charts, flow diagrams, model architecture diagrams and content formatting for presentations and reports.

Course Content

Course Code: 25ECAC307	Course Title: Generative AI	
L-T-P: 3-0-1	Credits: 4	Contact Hrs.: 70
ISA Marks: 50	ESA Marks: 50	Total Marks: 100
Teaching Hrs: 42	Lab Hrs:28	ESA Theory Exam Duration: 3 hrs.

Unit - 1		
1	Introduction to NLP Introduction to Natural Language Processing, Applications of Natural Language Processing, Word embeddings. Parsing techniques - Dependency Grammar, Neural dependency parsing.	06 hrs
2	Sequence-to-Sequence and Transformer Models: Machine Translation, Seq2Seq architecture, Attention mechanisms, Autoencoders/Decoders, Transformer architecture for text generation.	06 hrs
3	Generative Adversarial Networks Generative vs. Discriminative models, Generative Adversarial Networks and Language Models, types of GANs.	05 hrs
Unit - 2		
4	Large Language Models Introductions to LLM's, LLM - BERT and GPT models, prompting techniques, Adapters and low rank adoption (LoRA).	07 hrs
5	Diffusion models Diffusion models – continuous vs discrete, deterministic vs stochastic models.	05 hrs
6	Vision Language models and Multimodal Generative AI: Joint vision–language representation learning, CLIP – style contrastive learning, Text-to-Image (Stable Diffusion, DALL·E)	06 hrs
Unit - 3		
7	Retrieval-Augmented Generation (RAG): Architecture, embeddings, vector databases, context retrieval, and integration using frameworks, Different types of RAG	8 hrs

Text Books:

- 1 Daniel Jurafsky and James H. Martin, *Speech and Language Processing*, 3rd ed. (Draft), Stanford, CA, USA: Stanford University, 2023.
- 2 Ian Goodfellow, Yoshua Bengio, and Aaron Courville, *Deep Learning*, Cambridge, MA, USA: MIT Press, 2016

References:

- 1 A. Vaswani et al., "Attention Is All You Need," in *Proc. NeurIPS*, 2017.
- 2 Y. Goldberg, *Neural Network Methods for Natural Language Processing*. San Rafael, CA, USA: Morgan & Claypool Publishers, 2017

Evaluation Scheme (ISA)

Assessment	QP Max Marks	Weightage in Marks
In Semester Assessment-1 (ISA 1)	40	37
In Semester Assessment-2 (ISA 2)	40	
Lab Assessment	40	13
Total		50

Evaluation Scheme (ESA)

Assessment	Conducted for Marks	Weightage in Marks
ESA (Theory)	100	37
ESA (Lab)	20	13
Total		50

Minimum 20 marks in ISA for eligibility

In-Semester Assessment Scheme
Course Unitization for Minor Exams and End Semester Assessment

Topics / Chapters	Teaching Credits	No. of Questions in Minor Exam-1	No. of Questions in Minor Exam-2	No. of Questions in Activity	No. of Questions in ESA
Unit I					
Introduction to NLP	6	1	-	-	1
Sequence-to-Sequence and Transformer Models	6	1	-	-	1
Generative Adversarial Networks	5	1	-	-	1
Unit II					
Large Language Models	7	-	1	-	1
Diffusion models	5	-	1	-	1
Vision Language models and Multimodal Generative AI	6	-	1	-	1

Note

- Each Question carries 20 marks and may consist of sub-questions.
- Mixing of sub-questions from different chapters within a unit (only for Unit I and Unit II) is allowed in Minor I, II and ESA
- Answer 4 full questions of 20 marks each (two full questions from Unit I, II) 1 full question from Unit III in ESA.

Date:

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Course Assessment Plan

Course Title: Generative AI		Code: 25ECAC307				
Course outcomes (COs)	Weightage in assessment	Assessment Methods				
		ISA1	ISA2	Lab activity and course project	ESA-Lab exam	ESA-Theory Exam
1. Apply NLP techniques such as word embeddings and dependency parsing for text understanding.	10%	✓		✓	✓	✓
2. Implement Seq2Seq architecture, Attention, and transformer models for machine translation and text generation.	10%	✓		✓	✓	✓
3. Apply generative models such as GANs and diffusion models for language and image generation tasks.	20%	✓		✓	✓	✓
4. Adapt and evaluate Large Language Models and multimodal models for NLP and vision-language applications.	20%		✓	✓	✓	✓
5. Implement Retrieval-Augmented Generation (RAG) using embeddings, vector databases, and retrieval frameworks.	20%		✓	✓	✓	✓
6. Develop a comprehensive project report and present research outcomes suitable for publication in a technical conference.	20%			✓	✓	
Weightage		37%		13%	50%	

GenAI Lab Plan:

List of Experiments:

Topic	Experiment Title	Learning Outcome
1. Introduction to NLP	Text preprocessing: tokenization stop word removal stemming lemmatization	Understand NLP preprocessing pipeline
2. Word Embeddings	Implement Word2Vec (CBOW and Skip-Gram) and visualize embeddings	Learn semantic representation of words
3. Parsing Techniques	Dependency parsing and visualization of parse trees	Understand syntactic structure and dependency grammar
4. Seq2Seq Models	Implement Seq2Seq model for machine translation	Learn encoder-decoder architecture
5. Transformer Models	Text generation using pretrained Transformer (GPT-2)	Understand transformer-based text generation
6. Generative Adversarial Networks	Implement simple GAN for synthetic data generation	Understand generator and discriminator training
7. Large Language Models	Prompt engineering and fine-tuning using LoRA	Learn LLM adaptation and prompting techniques
8. Diffusion Models	Generate images using Stable Diffusion	Understand diffusion-based generative models
9. Vision Language Models	Image captioning using CLIP model	Learn multimodal vision-language integration
10. RAG	Build RAG pipeline using vector database and embeddings	Understand context-aware retrieval augmented generation

Chapter 1: Introduction to NLP

Lab 1: Introduction to NLP and its Applications Objective: Understand the basics of Natural Language Processing (NLP) and explore real-world applications.	Experiments: - Write a short Python script to extract keywords from a given text. - Explore NLP applications in fields like healthcare, finance, and chatbots. - Explore NLP datasets like 20 newsgroups, IMDB, Stanford Sentiment Treebank etc.,
	Tools: Python, NLTK, spaCy
Lab2: WordEmbeddings-Word2Vec, GloVe, and FastText Objective: Learn how word embeddings represent words as numerical vectors.	Experiments: - Implement Word2Vec using Gensim. - Load pre-trained GloVe embeddings and analyze word similarities. - Use Fast Text for out-of-vocabulary word representations.
	Tools: Gensim, spaCy, pre-trained embeddings
Lab 3: Parsing Techniques - Dependency Grammar and Named Entity Recognition Objective: Understand how dependency parsing and Named Entity Recognition (NER) work.	Experiments: - Perform dependency parsing using spaCy. - Identify named entities in a text.
	Tools: spaCy
Chapter 2: Sequence-to-Sequence and Transformer Models	
Lab4: Machine Translation with Seq2Seq and Attention Mechanisms	Experiments: Use an open-source translation model (e.g.,

Objective: Understand sequence-to-sequence models for language translation.	's Transformers) to translate text. 2. Implement a basic Seq2Seq model using TensorFlow.
	Tools: Transformers, Pytorch
Chapter 3: Advanced NLP Models	
Lab 5: Introduction to Generative Adversarial Networks (GANs) in NLP Objective: Understand GANs and their application in text generation.	Experiments: 1. Train a simple text-generating GAN using TensorFlow. 2. Compare Generative vs. Discriminative models.
	Tools: PyTorch
Lab6: Transformer Networks for Text Generation Objective: Implement transformer models like GPT for text generation.	Experiments: 1. Generate text using OpenAI's GPT model. 2. Fine tune a transformer model for text completion.
	Tools: Transformers, OpenAI API
Lab7: Exploring BERT and GPT Models Objective: Learn how to use and fine-tune large language models.	Experiments: 1. Perform text classification using BERT. 2. Generate text using GPT-3.
	Tools: Transformers, OpenAI API
Lab8: Prompt Engineering and LoRA Adapters Objective: Learn how to use prompts and adapt large models using LoRA effectively.	Experiments: 1. Experiment with different prompting techniques. 2. Fine-tune a model with LoRA for task-specific NLP.
	Tools: LoRA adapters
Lab 9: Diffusion Models Objectives: Understand diffusion models	Experiments: 1. Load Stable Diffusion pretrained model 2. Generate images using text prompts 3. Modify prompts and analyze output variations
	Tools : Python
Lab 9: Vision Language Models Objectives: Learn multimodal integration	Experiments: 1. Load pretrained CLIP model 2. Perform image-text similarity matching 3. Generate captions for images
	Tools: OpenAI CLIP, PyTorch
Lab 10: RAG Objectives: Understand RAG systems	Experiments: 1. Generate embeddings using sentence transformers 2. Store and retrieve vectors using FAISS/ChromaDB 3. Build RAG pipeline using LangChain
	Tools: LangChain, FAISS/ChromaDB

Lab Evaluation Rubrics:

Chapter	Criteria	Good (4-5 Marks)	Average (2-3 Marks)	Poor (0-1 Marks)	PI Code	CO
Introduction to NLP	Accuracy and correctness of NLP techniques	Complete implementation with accurate outputs and minimal errors	Partial implementation with some inaccuracies	Incorrect implementation with meaningless output	1.3.2	CO1
	Code efficiency and optimization	Optimized code handling large datasets efficiently	Moderately optimized code	Inefficient and slow implementation	2.2.2	CO1
	Code structure and documentation	Well-structured readable documented code	Moderate readability and structure	Poor structure and unreadable code	12.1.2	CO1
Seq2Seq and Attention	Correctness of model implementation	Correct encoder-decoder implementation	Partial implementation with minor errors	Incorrect or non-functional model	13.1.1	CO2
	Performance evaluation	Proper evaluation using metrics	Limited evaluation	No evaluation performed	13.1.1	CO2
	Result interpretation	Clear interpretation and comparison	Partial analysis	No interpretation	1.3.2	CO2
Generative Adversarial Networks	GAN implementation correctness	Correct and stable GAN training	Partially correct implementation	Incorrect GAN implementation	13.1.1	CO3
	Model performance	Proper loss balancing and convergence	Moderate performance	Poor or unstable training	13.1.1	CO3
	Result interpretation	Clear visualization and explanation	Limited interpretation	No explanation	1.3.2	CO3
Transformer Models	Correct implementation and optimization	Fully functional and optimized transformer	Partially correct implementation	Incorrect implementation	13.1.1	CO4
	Performance evaluation	Proper evaluation using metrics	Limited evaluation	No evaluation	13.1.1	CO4
	Architecture understanding	Clear explanation of transformer components	Partial explanation	Poor understanding	12.1.2	CO4

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Diffusion Models	Correct implementation and generation	Stable and accurate output generation	Moderate performance	Incorrect implementation	13.1.1	CO3
	Result evaluation	Proper evaluation using metrics	Partial evaluation	No evaluation	13.1.1	CO3
	Output interpretation	Clear explanation of generation process	Partial explanation	Poor understanding	12.1.2	CO3
Vision Language Models	Multimodal model implementation	Fully functional multimodal system	Partial implementation	Incorrect implementation	13.1.1	CO5
	Performance evaluation	Proper evaluation using similarity metrics	Limited evaluation	No evaluation	2.2.2	CO5
	Architecture explanation	Clear explanation of CLIP and diffusion architecture	Partial explanation	Poor explanation	12.1.2	CO5
Retrieval-Augmented Generation	RAG pipeline implementation	Fully functional RAG pipeline	Partial implementation	Incorrect implementation	13.1.1	CO5
	Retrieval performance	Accurate context retrieval	Moderate performance	Poor retrieval performance	2.2.2	CO5
	System analysis	Clear explanation and evaluation	Partial explanation	Poor explanation	12.1.2	CO5

Course Project Rubrics:					
Parameter	Good (4M – 5M)	Average (2M-3M)	Poor (0M – 1M)	PI	CO
Problem statement identification (Review-1)	Finalization of the problem statement and domain knowledge is well understood with an exhaustive literature survey related to the problem statement is performed.	Problem statement is identified, with poor domain understanding and proper literature survey not carried out.	Problem statement is identified and no domain understanding and no literature survey.	2.1.1	6
Parameter	Good (4M – 5M)	Average (2M-3M)	Poor (0M – 1M)	PI	CO
Assessing the Implementation (Review-2)	Complete implementation of the course project along with accurate model performance results.	80% - 90% implementation of the course project along with moderate model performance results.	60%-70% implementation of the course project with poor model performance results.	13.1.1	6
Parameter	Good (4M – 5M)	Average (2M-3M)	Poor (0M – 1M)	PI	CO
Prepare the report/paper, incorporating the Introduction, Literature Review, and Methodology (Review-3)	5-page double column report according to IEEE standard is written with $\leq 15\%$ plagiarism similarity score.	5-page double column report according to IEEE standard is written with $\leq 25\%$ plagiarism similarity score.	5-page double column report according to IEEE standard is written with $\leq 40\%$ plagiarism similarity score.	9.1.2	6
Parameter	Good (4M – 5M)	Average (2M-3M)	Poor (0M – 1M)	PI	CO
Incorporate the results, conclusion, and future scope of the report/paper (Review-4)	Results containing accurate model performance along with future scopes.	Results containing moderate model performance along with future scopes.	Results containing poor model performance.	10.1.4	6
Parameter	Good (4M – 5M)	Average (2M-3M)	Poor (0M – 1M)	PI	CO
The demonstration of the course project (Review-5)	Model is completely working for all the test cases and with good training and testing performance.	Model is working for few test cases and with moderate training and testing performance.	Model is not working for test cases and with poor training and testing performance.	8.1.2	6
Parameter	Good (4M – 5M)	Average (2M-3M)	Poor (0M – 1M)	PI	CO

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Paper submission mail, complete report/paper with appendices and a minimum 3-minute video about the project work carried out, verified by the guide. (Review-5)	The submission mail received from the conference/journal communicated for the paper/report. Video is well scripted, edited and presented.	The submission mail received from the conference communicated for the paper/report. Video is not well scripted, edited and presented.	No communication from the publisher received. Video is poorly scripted, edited and presented.	9.1.2	6
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Chapter wise Plan

Course Code and Title: 25ECAC307/ Generative AI	
Chapter Number and Title: 1. Introduction to NLP	Planned Hours:4 hrs

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PI Code
1. Understand the fundamentals and applications of Natural Language Processing (NLP)	1	L2	12.1.2
2.Choose the appropriate techniques for vector space representation, cosine similarity in word embeddings.	1	L3	1.3.2
3.Perform word embeddings using models such as word2vec with cbow and skip gram techniques.	1	L3	2.2.2
4.Apply different Word embeddings techniques to improve performance in NLP tasks like classification and translation.	1	L3	2.2.2
5.Analyze sentence structure using parsing techniques	1	L3	1.3.2

Lesson Schedule
Class No. - Portion covered per hour
1.Introduction to Natural Language Processing, Applications of Natural Language Processing
2.Word embeddings –vector representation techniques such as OHE
3.Word embeddings –vector representation techniques such as BoW, TFIDFs
4. Word embedding models- Word2vec
5. Word embedding models- glove and fast text
6. Parsing techniques - Dependency Grammar, Neural dependency parsing

Review Questions

S.NO	Sl.No. - Questions	TLOs	BL	PI Code
1	Illustrate with an example the different types of tokenization	1	L2	1.3.2
2	Explain why stemming and lemmatization are both important for text processing in NLP. How do they differ in their approach?	1	L2	1.3.2
3	Imagine you're building a sentiment analysis system for social media posts. Describe the steps you would take in preprocessing the data, including tokenization, stop word removal, and either stemming or lemmatization.	2	L2	1.3.2
4	Describe the difference between the Continuous Bag of Words (CBOW) and Skip-Gram models in Word2Vec.	2	L2	2.2.2
5	How can the given text document be transformed into vectors using One-Hot Encoding, Bag of Words and TF-IDF.	2	L3	1.3.2
	Document Text Output			
	S1 He is a good boy 1			

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	S2	She is a good girl	1			
	S3	Boy and girl are good	1			
6	Considering the sentence given "iNeuron Company Is Related to Data Science", explain the working of CBOW and Skip-Gram models in Word2Vec.			2	L3	2.2.2
7	Explain how a Dependency Tree represents sentence structure. What does each node and edge signify?			3	L2	1.3.2
8	Given a sentence, how would you use a Dependency Tree to identify the grammatical structure and relationships between words? Walk through the process of creating a Dependency Tree manually.			3	L3	1.3.2
9	Why is POS tagging important in Natural Language Processing? What role does it play in understanding the syntactic structure of a sentence?			4	L2	1.3.2
10	Imagine you're designing a chatbot that needs to understand locations and dates mentioned by users. How would you use both POS tagging and NER to improve its understanding of user input?			4	L3	1.3.2

Course Code and Title: 25ECAC307/ Generative AI

Chapter Number and Title: 2. Transformer Models:

Planned Hours: 6hrs

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PI Code
1. Understand the fundamentals of Machine Translation, its evolution, key approaches, and real-world applications in NLP.	2	L2	12.1.2
2. Understand the types of machine translation approaches	2	L2	2.2.3
3. Explain the working of encoders and decoders with their architecture	2	L2	13.1.1
4. Use Seq2Seq architecture and their applications in NLP tasks such as machine translation and text summarization	2	L3	13.1.1
5. Explain the working principles and architecture of the Attention mechanism in NLP tasks such as machine translation and text summarization	2	L2	13.1.1

Lesson Schedule

Class No. - Portion covered per hour

1. Introduction to Machine Translation

2. Seq2Seq: fundamental architecture

3. Types of Translation techniques: Rule-Based Machine Translation, Statistical Machine Translation, Neural Machine Translation

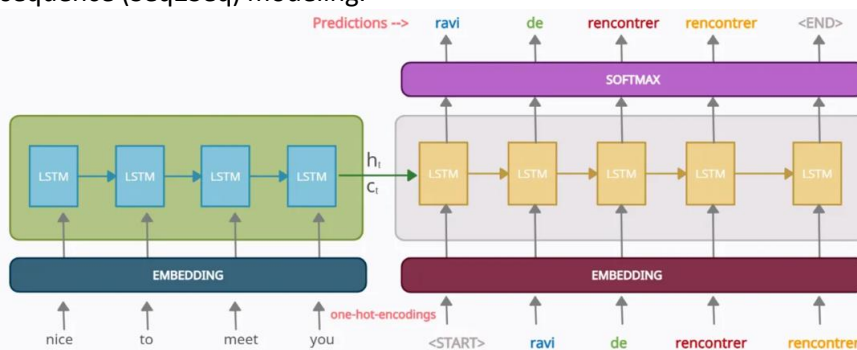
4. Attention: Working principles, Architecture

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5. Autoencoder: Introduction, Architecture of Autoencoder, Types of Autoencoders

6. Decoders: Introduction, Architecture of decoder

Review Questions

Sl.No	Questions	TLOs	BL	PI Code												
1	What is Machine Translation (MT)? What are the primary steps of MT in Natural Language Processing (NLP)?	1	L2	12.1.2												
2	Is it possible to use Artificial Neural Networks (ANN) to process and analyze sequential data in Natural Language Processing (NLP)? Illustrate with the help of an example.	2	L3	12.1.2												
3	How does forward and backward propagation work in a Recurrent Neural Network (RNN) for the given dataset? Provide a detailed explanation with necessary equations. <table border="1"><thead><tr><th>Document</th><th>Text</th><th>Output</th></tr></thead><tbody><tr><td>S1</td><td>The food is good</td><td>1</td></tr><tr><td>S2</td><td>The food is bad</td><td>0</td></tr><tr><td>S3</td><td>The food is not good</td><td>0</td></tr></tbody></table>	Document	Text	Output	S1	The food is good	1	S2	The food is bad	0	S3	The food is not good	0	2	L3	13.1.1
Document	Text	Output														
S1	The food is good	1														
S2	The food is bad	0														
S3	The food is not good	0														
4	What are the key components of a Seq2Seq model?How does a Seq2Seq model differ from a standard RNN.	3	L3	13.1.1												
5	How does an LSTM store long-term dependencies in a sequence?	3	L2	12.1.2												
6	Analyze the given figure and explain the concept of sequence-to-sequence (Seq2Seq) modeling. 	3	L3	13.1.1												
7	How does attention mechanism address the limitations of Seq2Seq models, particularly the fixed-length context vector?	4	L2	13.1.1												
8	You are tasked with improving the performance of a Seq2Seq machine translation model that uses a fixed-length context vector. How would you integrate an attention mechanism to enhance the translation accuracy? Walk through the steps of adding attention to the model.	4	L3	13.1.1												
9	How does an autoencoder learn to reconstruct data?	4	L3	13.1.1												
10	What is a variational autoencoder (VAE), and how does its architecture differ?	5	L2	13.1.1												

Course Code and Title: 25ECAC307/ Generative AI	
Chapter Number and Title: 3. Generative Adversarial Networks	Planned Hours: 5hrs

Learning Outcomes:

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PI Code
1.Understand the Principles and Architecture of Generative model.	3	L2	2.2.3
2. Understand the Principles and Architecture of Discriminative model	3	L2	2.2.3
3. Differentiate Generative and Discriminative models	3	L3	13.1.1
4.Choose one of the types of GAN for a particular task.	3	L3	13.1.1
5.Explain the types of Language Models like Pure Statistical Methods, Neural Models	3	L2	13.1.1

Lesson Schedule
Class No. - Portion covered per hour
1. Generative model and Discriminative model
2. Generative vs. Discriminative models
3. Generative Adversarial Networks
4. Types of GAN
5. Language Models: Pure Statistical Methods, Neural Models

Review Questions

Sl.No.	Questions	TLOs	BL	PI Code
1	Explain the architecture of Generative Model. What are the key differences between generative and discriminative models?	1	L2	2.2.3
2	How does a generative model learn the data distribution? Why are discriminative models more efficient for classification tasks?	2	L2	13.1.1
3	What type of GAN would you use for realistic image generation, and why?	3	L3	13.1.1
4	What is a discriminative model? Which loss functions are commonly used in discriminative models?	3	L2	2.2.3
5	Differentiate between Discriminative and Generative models with an example.	3	L3	13.1.1
6	Can a model be both generative and discriminative? Give examples. What type of GAN would you use for converting satellite images to street maps?	4	L3	13.1.1

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Course Code and Title: 25ECAC307/ Generative AI	
Chapter Number and Title: 4. Large Language Models	Planned Hours: 8hrs

Learning Outcomes:

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PI Code
1. Understand LLM's capabilities and applications	4	L2	1.3.2
2. Explain the working principles of BERT's model architecture, contextual word representations and Next-sentence Tasks	4	L2	2.2.2
3. Explain working principles of GPT, autoregressive nature, and applications in text generation, conversation.	4	L2	2.2.2
4. Differentiate BERT's model and GPT model	4	L3	2.2.2
5. Choose one of the types of prompting techniques for a specific task	4	L3	2.2.2
6. Explain how to generate models and adapters in formats	4	L2	2.2.2

Lesson Schedule
Class No. - Portion covered per hour
1. Introductions to LLM's
2. LLM – BERT model
3. LLM - GPT model
4. prompting techniques
5. prompting techniques
6. Adapters
7. low rank adoption (LoRA)

Review Questions

Sl.No. - Questions	TLOs	BL	PI Code
Q1. What is a Large Language Model (LLM), and how does it function? Discuss its key characteristics and illustrate with examples some real-world applications across different domains."	1	L2	1.3.2
Q2. Explain the concept of prompting in Large Language Models?	2	L2	2.2.2
Q3. How do different prompting techniques (zero-shot, few-shot, and chain-of-thought prompting) affect the model's output and performance?	3	L2	2.2.2
Q4. What are adapters in the context of Large Language Models?	4	L2	2.2.2
Q5. Describe the Low-Rank Adaptation (LoRA) technique for fine-tuning Large Language Models.	2	L3	2.2.2
Q6. How does LoRA reduce computational costs while maintaining model performance?"	5	L2	2.2.2
Q7. How does LoRA balance parameter efficiency and model performance, and what are its advantages and potential limitations compared to traditional full fine-tuning?"	6	L2	2.2.2

Course Code and Title: 25ECAC307/ Generative AI	
Chapter Number and Title: 5. Diffusion models	Planned Hours:07 hrs

Learning Outcomes:
At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PI Code
1.Explain the working principles of Diffusion model	4	L2	13.1.1
2.Choose the type of Diffusion models for a given NLP task	4	L3	1.4.4
3. Differentiate Deterministic models and stochastic models	4	L3	13.1.1
4. Differentiate continuous and discrete models	4	L3	13.1.1
5. Differentiate Linear and non-Linear model	4	L3	13.1.1

Lesson Schedule
Class No. - Portion covered per hour
1. Transformer Networks, transformers for text generation
3. Diffusion models: discrete
4. Continuous vs discrete
5. Deterministic models and stochastic models
6. Deterministic vs stochastic models

Review Questions

Sl.No. - Questions	TLOs	BL	PI Code
Q1. Why are Transformer models generally preferred over RNNs and LSTMs for natural language processing tasks?	1	L2	13.1.1
Q2. What are the key features of Transformer models that make them effective for NLP tasks?	1	L2	13.1.1
Q3. Analyse the working of the Transformer encoder-decoder architecture with the help of a concrete example, highlighting how information flows from input to output through self-attention, cross-attention, and feed-forward layers..	1	L2	13.1.1
Q4. How does Multi-Head Attention it enhance the model's ability to capture different aspects of input relationships? Illustrate your answer with an example.	2	L3	1.3.2
Q5. Explain the architecture and training strategy of BERT (Bidirectional Encoder Representations from Transformers). How does its bidirectional attention mechanism differ from traditional left-to-right models, and what advantages does this offer for downstream NLP tasks?	4	L2	13.1.1
Q6. Explain the architecture and training methodology of Generative Pretrained Transformers (GPT). How does its autoregressive nature influence its performance in text generation tasks.	3	L2	13.1.1
Q7. Compare and contrast BERT and GPT in terms of their architectures, training objectives, and applications.	3	L2	13.1.1
Q8. Analyze the working principles of diffusion models in generative modeling. How does the iterative denoising process contribute to high-	5	L2	13.1.1

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quality sample generation, and how do diffusion models compare to GANs in terms of training stability and output diversity			
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Course Code and Title: 25ECAC307/ Generative AI	
Chapter Number and Title: 6. Vision Language Models and Multimodal Generative AI	Planned Hours: 6hrs

Learning Outcomes:

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PI Code
1. Explain the fundamentals of multimodal AI and the importance of combining vision and language modalities.	5	L2	1.3.2
2. Describe joint vision–language representation learning and its role in multimodal understanding.	5	L2	13.1.1
3. Explain the architecture and working principles of CLIP and contrastive learning.	5	L2	13.1.1
4. Analyze how contrastive learning aligns visual and textual embeddings in a shared feature space.	5	L3	2.2.2
5. Explain the working principles of text-to-image generation models such as Stable Diffusion and DALL·E.	5	L2	13.1.1
6. Apply multimodal generative AI models for applications such as image generation, captioning, and cross-modal retrieval.	5	L3	12.1.2

Lesson Schedule
Class No. - Portion covered per hour
1. Introduction to Multimodal AI, Vision Language Models, need for multimodal learning
2. Joint Vision–Language Representation Learning, embedding alignment, multimodal feature learning
3. CLIP Architecture: Image encoder, Text encoder, embedding space
4. Contrastive Learning: Working principles, loss function, training mechanism
5. Text-to-Image Generation: Stable Diffusion architecture, latent diffusion models
6. DALL·E architecture, multimodal generative applications, real-world use cases

Review Questions

Sl.No. - Questions	TLOs	BL	PI Code
1.What is Multimodal AI? Explain the importance of integrating vision and language in AI systems.	1	L2	1.3.2
2.Describe joint vision–language representation learning. How does it help improve multimodal understanding?	2	L2	13.1.1

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3. Explain the architecture of CLIP. How do the image encoder and text encoder generate aligned embeddings?	3	L2	13.1.1
4. How does contrastive learning work in CLIP? Explain how it aligns image and text representations in the same feature space.	4	L3	2.2.2
5. Explain how CLIP performs image-to-text and text-to-image similarity matching with an example.	4	L3	2.2.2
6. Describe the architecture and working mechanism of Stable Diffusion for text-to-image generation.	5	L2	13.1.1
7. Compare Stable Diffusion and DALL·E in terms of architecture, training approach, and applications.	5	L3	13.1.1
8. Design a multimodal AI system using vision-language models for image captioning or visual search. Explain its workflow.	6	L3	12.1.2

Course Code and Title: 25ECAC307/ Generative AI	
Chapter Number and Title: 7. Retrieval-Augmented Generation	Planned Hours: 8hrs

Learning Outcomes:

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PI Code
1. Explain the need for Retrieval-Augmented Generation (RAG) and its role in overcoming limitations of standalone Large Language Models.	5	L2	13.1.1
2. Describe the architecture and components of RAG including embedding model, retriever, vector database, and generator.	5	L2	12.1.2
3. Explain the concept of embeddings and semantic similarity for efficient information retrieval.	5	L2	2.2.2
4. Apply vector databases such as FAISS or ChromaDB for storing and retrieving embedding vectors.	5	L3	12.1.2
5. Analyze context retrieval mechanisms such as similarity search and top-k retrieval to improve response quality.	5	L3	1.3.2
6. Implement a basic RAG pipeline using frameworks such as LangChain.	5	L3	13.1.1
7. Compare different types of RAG architectures such as Naïve RAG, Advanced RAG, and Hybrid RAG.	5	L3	2.2.3
8. Design a domain-specific knowledge-grounded AI system using RAG architecture.	5	L3	13.1.1

Lesson Schedule
Class No. - Portion covered per hour
1. Introduction to Retrieval-Augmented Generation (RAG), need for external knowledge integration in LLMs
2. Architecture of RAG: Retriever, Vector Database, Embedding Model, Generator

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3. Embeddings: semantic vector representations, cosine similarity, embedding generation using pretrained models
4. Vector Databases: FAISS, ChromaDB, Pinecone – indexing, similarity search, scalability
5. Context Retrieval Mechanisms: top-k retrieval, similarity search, reranking
6. Integration Frameworks: LangChain, , OpenAI API for RAG pipeline
7. Types of RAG: Naïve RAG, Advanced RAG, Hybrid RAG, Graph-based RAG
8. Applications and Case Studies: Question Answering, Chatbots, Domain-specific AI systems

Review Questions

Sl.No.	Questions	TLOs	BL	PI Code
1	What is Retrieval-Augmented Generation (RAG)? Explain its importance in improving LLM performance.	1	L2	13.1.1
2	Explain the architecture of a RAG system with the help of a block diagram.	1	L2	13.1.1
3	What are embeddings? How do they help in semantic search and retrieval?	2	L2	2.2.2
4	Explain cosine similarity and its role in vector retrieval systems.	2	L3	2.2.2
5	Compare traditional keyword-based retrieval with embedding-based retrieval.	3	L3	1.3.2
6	How does a vector database such as FAISS improve retrieval efficiency? Explain with an example.	3	L3	12.1.2
7	Explain the process of context retrieval and how it improves response generation in RAG systems.	4	L3	13.1.1
8	Design a basic RAG pipeline for a domain-specific chatbot application. Explain each component.	5	L3	13.1.1
9	Differentiate between Naïve RAG, Advanced RAG, and Hybrid RAG.	6	L3	2.2.3
10	How can RAG be used in real-world applications such as healthcare, legal AI, or academic assistants?	6	L3	1.3.2

Model Question Paper for Minor Examination (ISA-I)						
Course Code: 25ECAC307			Course Title: Generative AI			
Duration: 75 mins			Max. Marks: 40			
Note: Answer any two full questions. Each full question carries equal marks.						
Q.No	Questions	Marks	CO	BL	PO	PI Code
1.a	Explain how the attention mechanism improves Seq2Seq models and overcomes fixed-length context vector limitations.	10	CO2	L2	1	1.3.2
b.	Differentiate between Generative and Discriminative models and explain GAN architecture.	10	CO3	L3	2	2.2.2
2.a	Explain the role of word embeddings and how CBOW and Skip-Gram capture semantic relationships.	10	CO1	L3	1	2.2.2
b.	Explain GAN training and how generator and discriminator compete.	10	CO3	L2	1	13.1.1
3.a	Describe Dependency Grammar and Neural Dependency Parsing with an example.	10	CO1	L2	1	1.3.2
b.	Explain Seq2Seq architecture and how encoder and decoder generate translated output.	10	CO2	L3	2	13.1.1

Model Question Paper for Minor Examination (ISA-II)						
Course Code: 25ECAC307			Course Title: Generative AI			
Duration: 75 mins			Max. Marks: 40			
Note: Answer any two full questions. Each full question carries equal marks.						
Q.No	Questions	Marks	CO	BL	PO	PI Code
Q1a	Explain the architecture and working principle of BERT and GPT models. Compare their bidirectional and autoregressive training approaches.	10	CO4	L3	2	2.2.2
Q1b	Explain continuous and discrete diffusion models. Differentiate between deterministic and stochastic diffusion processes.	10	CO3	L2	1	13.1.1
Q2a	Describe the working of CLIP and explain how contrastive learning aligns image and text embeddings.	10	CO5	L3	2	13.1.1
Q2b	Explain the concept of Low-Rank Adaptation (LoRA). How does it improve parameter efficiency in Large Language Models?	10	CO4	L2	1	2.2.2
Q3a	Explain the architecture of Stable Diffusion and its role in text-to-image generation.	10	CO5	L2	1	13.1.1
Q3b	Compare prompting techniques such as zero-shot, few-shot, and chain-of-thought prompting in LLMs.	10	CO4	L3	2	1.3.2

Model Question Paper for End Semester Assessment (ESA)						
Course Code: 25ECAC307				Course Title: Generative AI		
Marks: 100				Duration: 180 mins		
	Note: Answer any 2 full questions from each UNIT I and Unit II					
UNIT – I						
Q.No	Questions	Marks	CO	PI Code	PO	BL
1 a	What is the impact of tokenization and Part-of-Speech (POS) tagging on the performance of natural language processing tasks such as sentiment analysis or named entity recognition? For example, in analyzing restaurant reviews, how does tokenization handle expressions like 'not very good' or 'absolutely loved it,' and in what ways does POS tagging help differentiate between usages of words like 'love' as a verb versus a noun?	10	CO2	L2	1	1.3.2
b	Explain the main differences between LSTM and GRU models. In what situations might one be preferred over the other?	10	CO3	L3	2	2.2.2
2 a	Explain how the architecture and data reconstruction mechanism of a Variational Autoencoder (VAE) differ from that of a standard autoencoder. Additionally, how are autoencoders leveraged in practical applications such as image detection and handwritten text recognition?	10	CO1	L3	1	2.2.2
b	How does Neural Machine Translation (NMT) model the syntactic structure and semantic dependencies of complex, multi-clause sentences to ensure accurate and context-preserving translation?	10	CO3	L2	1	13.1.1
3 a	Is it possible for a model to function as both generative and discriminative? Which specific type of Generative Adversarial Network (GAN) would be most effective for converting satellite imagery into street maps, and why?	10	CO1	L2	1	1.3.2
b	How do the architectures of DCGANs (Deep Convolutional GANs) differ from those of traditional GANs, and what advantages do DCGANs offer for image generation tasks?	10	CO2	L3	2	13.1.1
UNIT – II						
4 a	Analyze the differences between BERT and GPT architectures. How would you select the appropriate model for tasks such as question answering and text generation? Justify your answer.	10	CO4	L3	2	2.2.2
b	Design a prompt strategy using zero-shot, few-shot, or chain-of-thought prompting to improve performance of an LLM for a domain-specific task such as medical diagnosis or legal text analysis.	10	CO3	L2	1	13.1.1
5 a	Analyze how continuous and discrete diffusion models differ in their generation process. How would you select the appropriate diffusion model for image generation tasks?	10	CO5	L3	2	13.1.1

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b	Apply the concept of stochastic diffusion to generate high-quality images. Explain the denoising process step-by-step.	10	CO4	L2	1	2.2.2
6 a	Analyze the working of CLIP and explain how contrastive learning improves multimodal alignment between text and images. How can CLIP be applied for image retrieval tasks?	10	CO5	L2	1	13.1.1
b	Design a text-to-image generation workflow using Stable Diffusion or DALL·E. Explain how prompts, latent space, and diffusion process interact to generate the final image.	10	CO4	L3	2	1.3.2
UNIT III						
7a	Explain RAG architecture including retriever, vector database, and generator.	10	CO5	13.1.1	2	L2
7b	Explain embeddings and vector databases in RAG.	10	CO5	2.2.2	2	L3
8a	Design a RAG pipeline for chatbot applications.	10	CO5	12.1.2	1	L3
8b	Compare Naive, Advanced, and Hybrid RAG.	10	CO5	2.2.3	2	L3